discrete electron charge. Tunnelling cannot raise the electrostatic energy.

This is Coulomb blockade. Find the range of plate voltage

for which single-electron tunnelling is forbidden, that is, the blockade condition.

. If just permits tunnelling, estimate the capacitance .

. For Fig. at fixed voltage , use the double junction in Fig.

to avoid stray capacitance when observing tunnelling. The middle metal layer is

the single-electron island. The electrodes are . Its charge is , where is

any integer. is unit charge; capacitances are and .

Show that the charge-dependent (island) electrostatic energy is

. In Fig. , add a gate to the source , drain pair via capacitor .

The gate forms the single-electron transistor of Fig. . No single electron

Island

Island

Fig. Fig.

tunnels between gate and island. At small fixed , controls . How does

control ? Model as adding the equivalent island charge .

Its energy is about , with . On the grid (Fig.

), allow for blockade.

Start with . As

rises from to ,

draw the island's energy

against in bold.

Plot vertically; is in

Fig.